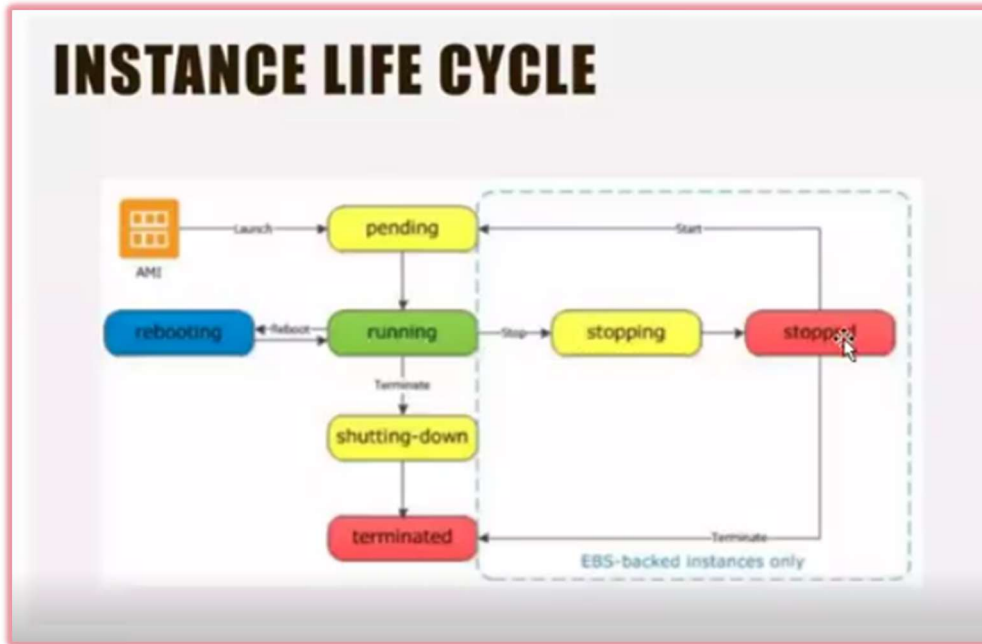


Aws – Jovac 2026

LECTURE 1

- Introduction to EC2
- Instance types explanation
- Scalability
- Instance Life Cycle
- Explore Aws MarketPlace



- Get instance screenshot (show present condition of instance)
- Key Pairs Explanation (key cant be changes after creating instance)
- AMI is stored in AWS and cost of storage is applied
- Intro to EBS
- Comparison of EBS vs Instance Store
 - Every server doesn't support Instance store

Storage Comparison

EBS Volume

An EBS volume is a network attached drive which results in slow performance but data is persistent meaning even if you reboot the instance data will be there.

Instance Store

An instance store is a physically attached device which gives better performance but data will be lost once instance is rebooted.

How to Calculate Costing

- Open <https://calculator.aws>
- Click create estimate

The screenshot shows the AWS Pricing Calculator homepage. At the top, there's a navigation bar with the AWS logo, 'pricing calculator', and links for 'Feedback', 'Language: English', 'Contact Sales', and 'Create an AWS Account'. The main heading is 'AWS Pricing Calculator' with the subtext 'Estimate the cost for your architecture solution'. Below this, it says 'Configure a cost estimate that fits your unique business or personal needs with AWS products and services.' On the right, there's a 'Create an estimate' section with a button 'Sign in for personalized estimates' and a 'Create estimate' button. At the bottom right, there are links to 'Log into the AWS console' and 'Discover the experience of AWS Pricing Calculator'.

- Select region
- Find service and select its configuration and specifications and usage

The screenshot shows the 'EC2 Instances (383)' section of the AWS Pricing Calculator. It displays the 'Chosen instance: t3a.2xlarge' with 'Family: t3a', '8vCPU', and '32 GiB Memory'. Below this, there's a search bar and filters for 'Instance family', 'vCPUs', 'Memory (GiB)', and 'Network performance'. A table lists various EC2 instance types with their specifications and costs. The 't3a.2xlarge' instance is selected as the lowest-cost option.

Instance name	vCPUs	Memory	Network Performance	Storage	On-Demand Hourly Cost	CurrentGeneration	Potential Effective Hourly Cost (Savings %)
t3a.2xlarge	8	32 GiB	Up to 5 Gigabit	EBS only	0.3443	Yes	0.2196 (36%)
t2.2xlarge	8	32 GiB	Moderate	EBS only	0.4588	Yes	0.2076 (55%)
t3.2xlarge	8	32 GiB	Up to 5 Gigabit	EBS only	0.5056	Yes	0.2789 (45%)
m5a.2xlarge	8	32 GiB	Up to 10 Gigabit	EBS only	0.59	Yes	0.4516 (23%)
m6a.2xlarge	8	32 GiB	Up to 12500 Megabit	EBS only	0.5902	Yes	0.4559 (23%)
m5ad.2xlarge	8	32 GiB	Up to 10 Gigabit	1 x 300 NVMe SSD	0.636	Yes	0.4689 (26%)
r5a.2xlarge	8	64 GiB	Up to 10 Gigabit	EBS only	0.654	Yes	0.4755 (27%)
r6a.2xlarge	8	64 GiB	Up to 12500 Megabit	EBS only	0.654	Yes	0.4809 (26%)
r5ad.2xlarge	8	64 GiB	Up to 10 Gigabit	1 x 300 NVMe SSD	0.7	Yes	0.4929 (30%)
m7i-flex.2xlarge	8	32 GiB	Up to 12500 Megabit	EBS only	0.7526	Yes	0.5091 (32%)

➤ Calculated costs shown

Instance: 0.3443/Hour

Monthly: 251.34/Month

▼ Show calculations

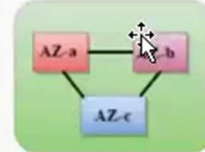
1 instances x 0.3443 USD On Demand hourly cost x 730 hours in a month = 251.339000 USD

On-Demand instances (monthly): 251.339000 USD

➤ Region and Availability Zones

Region

- A region is a geographical area. Each region consists of 2 more availability zones.
- A region is a collection of data centers which are completely isolated from other regions.
- A region consists of more than two availability zones connected to each other through links.

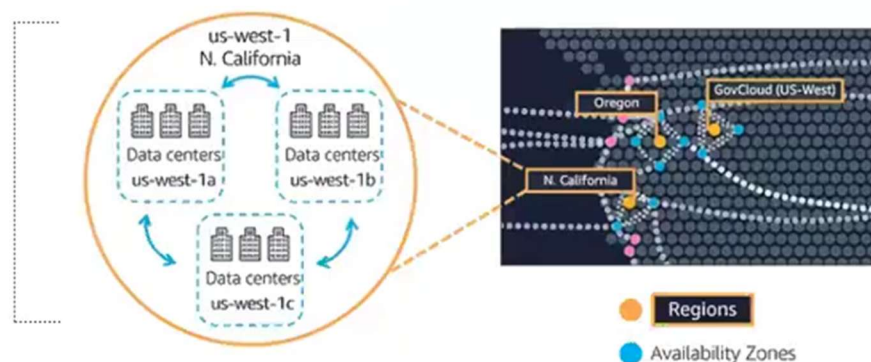


Availability zone as a Data Center

- An availability zone is a facility that can be somewhere in a country or in a city. Inside this facility, i.e., the Data Centre, we can have multiple servers, switches, load balancing, firewalls. The things that interact with the cloud sit inside the data centers.
- An availability zone can be several data centers, but if they are close together, they are counted as 1 availability zone.

I

Understanding AWS Availability Zones



➤ Ways to connect with AWS

Three ways to interact with AWS



AWS Management Console

Easy-to-use graphical interface



AWS Command Line Interface (AWS CLI)

Access to services by discrete command



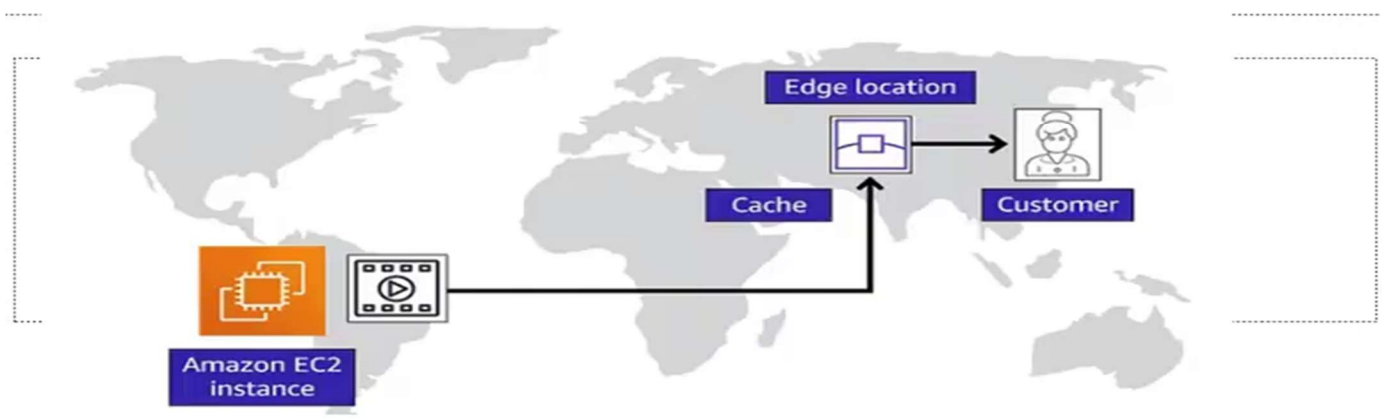
Software development kits (SDKs)

Access services in your code

➤ Edge Locations

Edge Locations

- Edge locations are the endpoints for AWS used for caching content.
- Edge locations consist of CloudFront, Amazon's Content Delivery Network (CDN).
- Edge locations are more than regions. Currently, there are over 150 edge locations.
- Edge location is not a region but a small location that AWS have. It is used for caching the content.
- Edge locations are mainly located in most of the major cities to distribute the content to end users with reduced latency.
- For example, some user accesses your website from Singapore; then this request would be redirected to the edge location closest to Singapore where cached data can be read.



AMI(Amazon Machine Image)

- An Amazon Machine Image (AMI) is a special type of virtual appliance that is used to create a virtual machine within the Amazon Elastic Compute Cloud (EC2).
- AMI is like an OS
- Like a Template (ready to create instance)
- An AMI is essentially a pre-configured template containing the necessary information to launch an instance in the AWS cloud. It can include the operating system, application server, applications, and related configurations. The introduction of AMIs significantly streamlines the process of server provisioning; allowing businesses to launch identical, consistent environments quickly and efficiently.

❖ AMI Architecture and Fundamentals

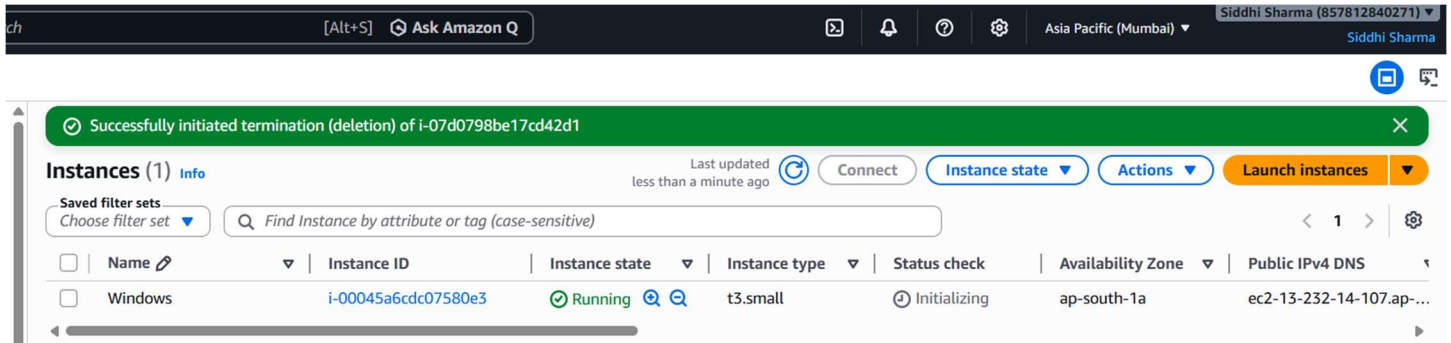
- At its core, an AMI is a combination of several essential components. Understanding these components is key to mastering AMIs:
- Operating System: The base software that runs on the instance, such as Amazon Linux, Ubuntu, or Windows.
- Application Server and Software: Includes any necessary application software, databases, or frameworks.
- Root Volume Snapshot: A point-in-time backup of the instance's root volume (i.e., the operating system and system configurations).
- Launch Permissions: These define who can launch an instance from the AMI. Permissions can be set for individual users or groups within AWS

❖ Types of Images

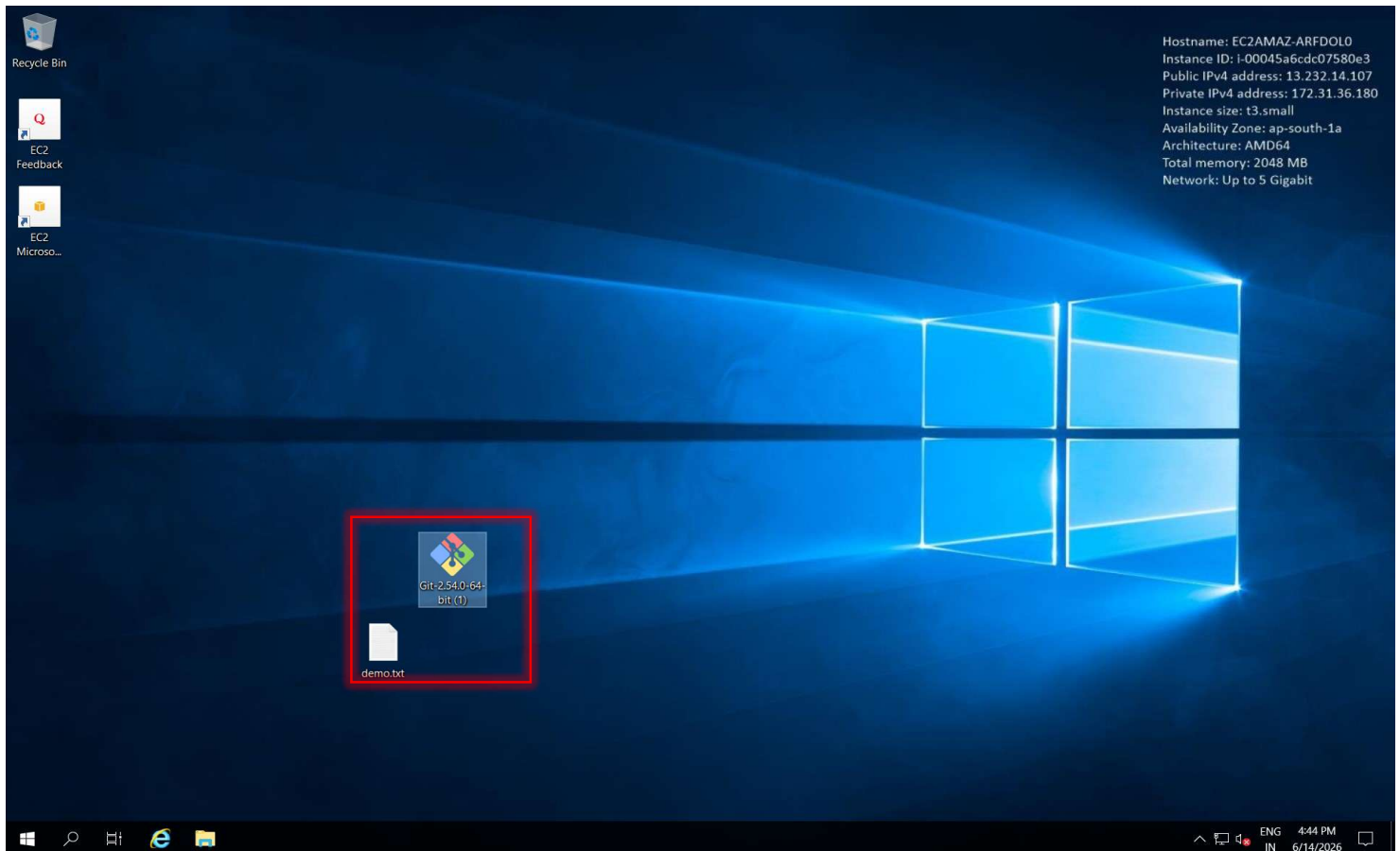
- Public: an AMI that can be used by anyone.
- Paid: a for-pay AMI that is registered with Amazon Marketplace and can be used by anyone who subscribes for it. Marketplace allows developers to mark-up Amazon's usage fees and optionally add monthly subscription fees.
- Shared: a private AMI that can only be used by Amazon EC2 users who are allowed access to it by the developer

AMI(Amazon Machine Image) - LAB

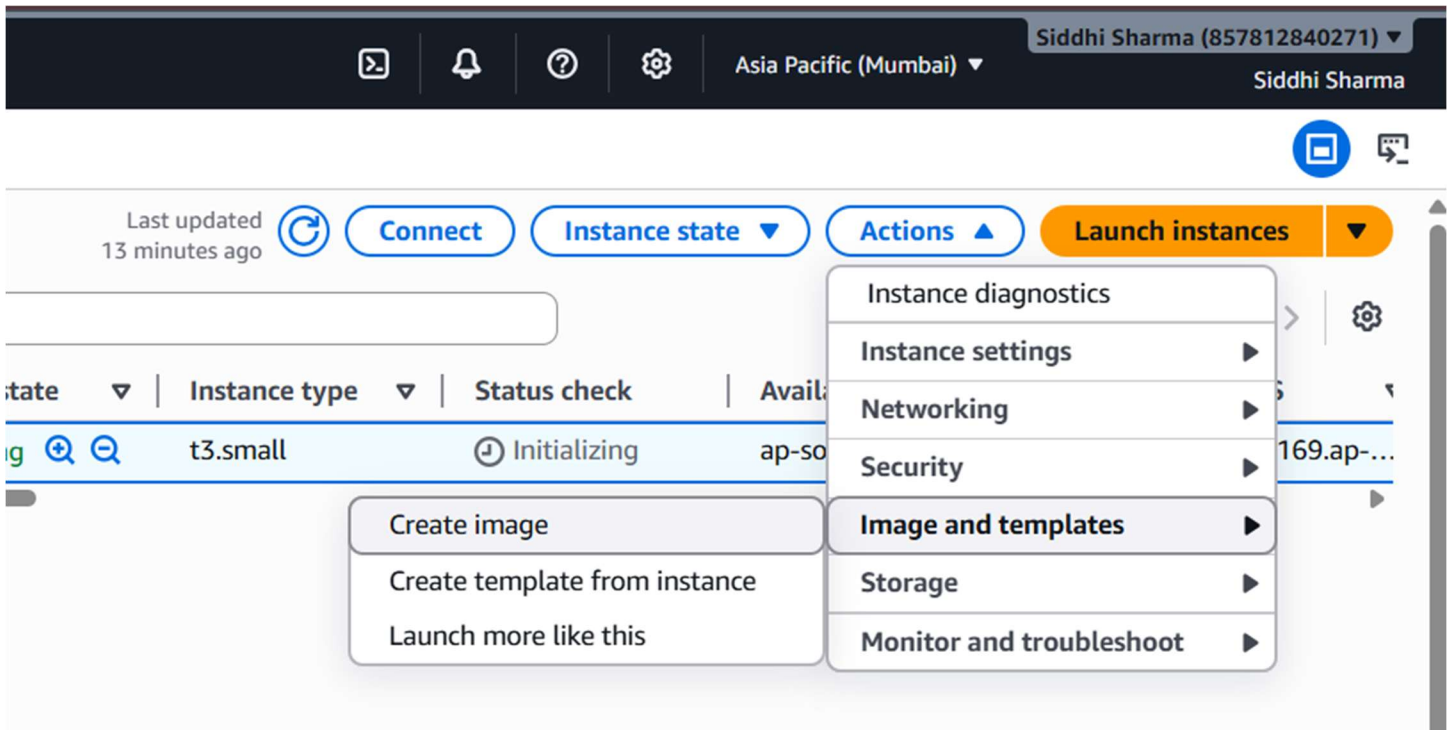
1. Create a new Windows Instance



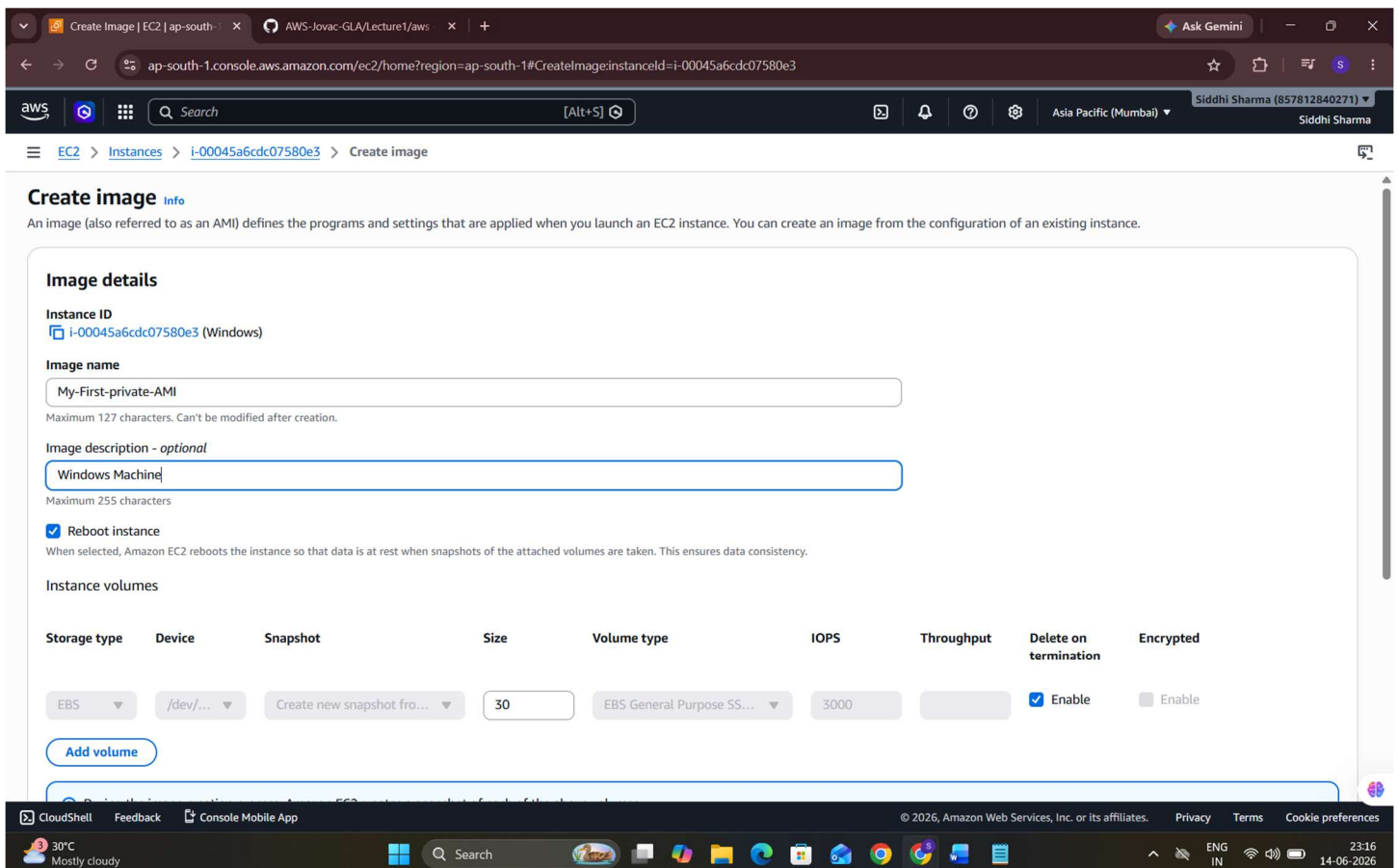
2. Access server using RDP client (Use steps in previous Lecture to get password and get access to server.
3. Enter ID Password
4. Create a TXT file in server
5. Install a Software(GITBASH) in server by copying setup to server



6. AWS suggest to stop the server whose you want to create AMI of.
7. Go to actions ->images and templates -> Create Image

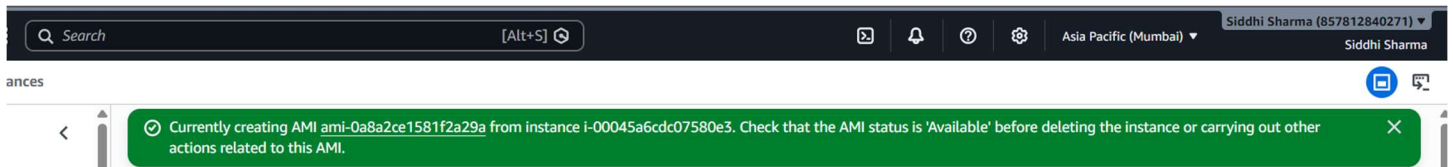


8. Create image window will open
9. Select AMI name

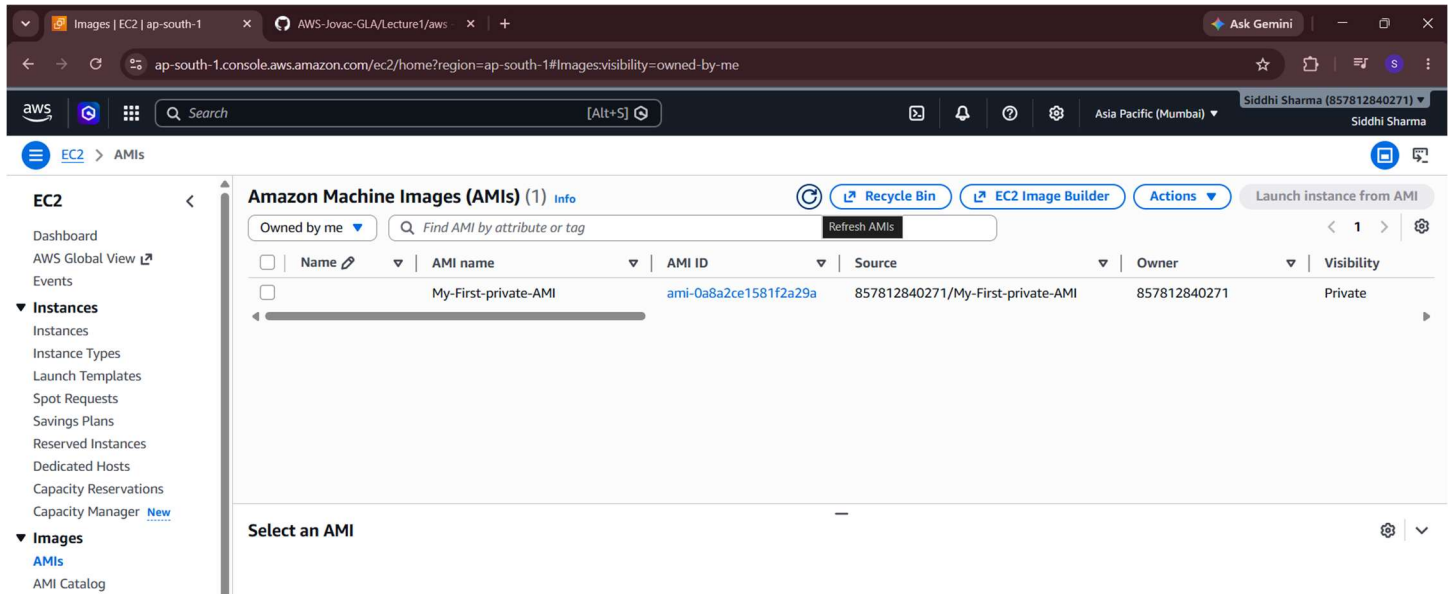


10. Check or uncheck Reboot option according to requirement

11. Create image par click kare to start creating



12. Ami created



13. Deregistration protection is about deleting AMI

- User can enable Termination protection by going to Action and then Instance Settings to enable or disable it

14. Go to EBS -> snapshots check percentage completed

